

The Digestive System

Small intestine

SGL2

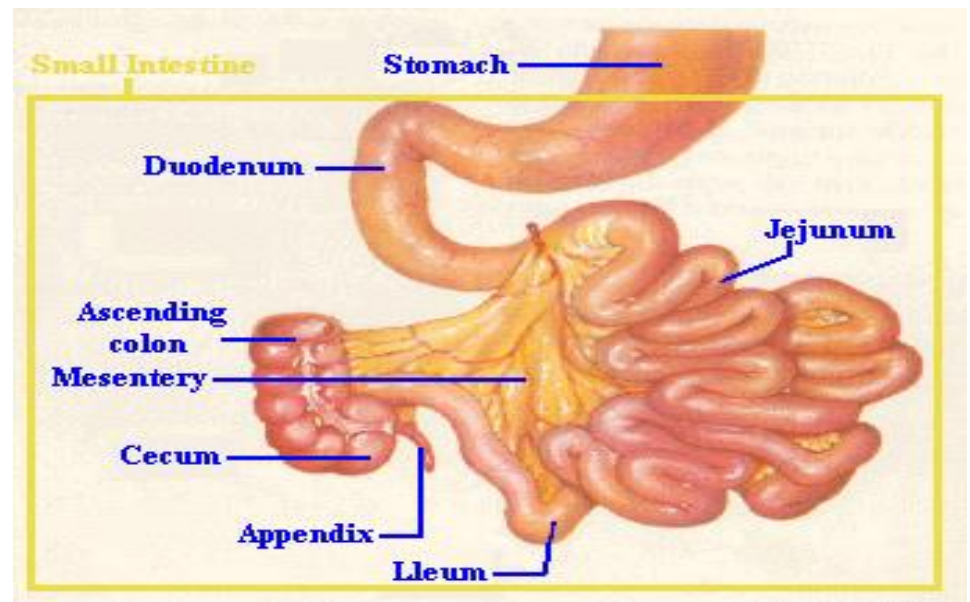
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Small Intestine

- The small intestine is relatively long—approximately 5 m this permitting prolonged contact between food & digestive enzyme consist of three regions :
 - duodenum 25cm
 - jejunum 2.5m
 - ileum 3.5m
 - Its distal end terminates at the ileocecal valve, a sphincter that controls the entry of materials into the large intestine

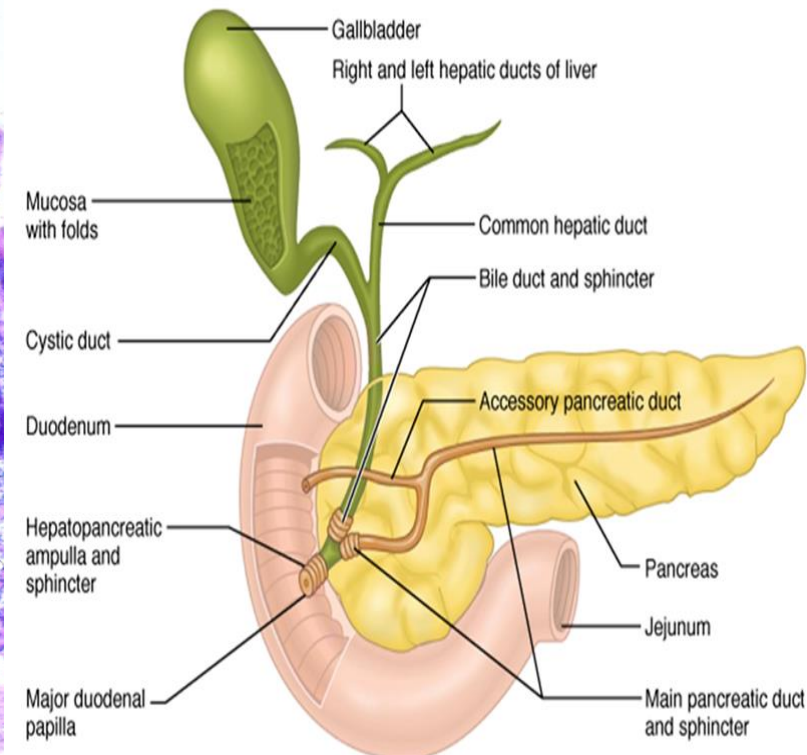
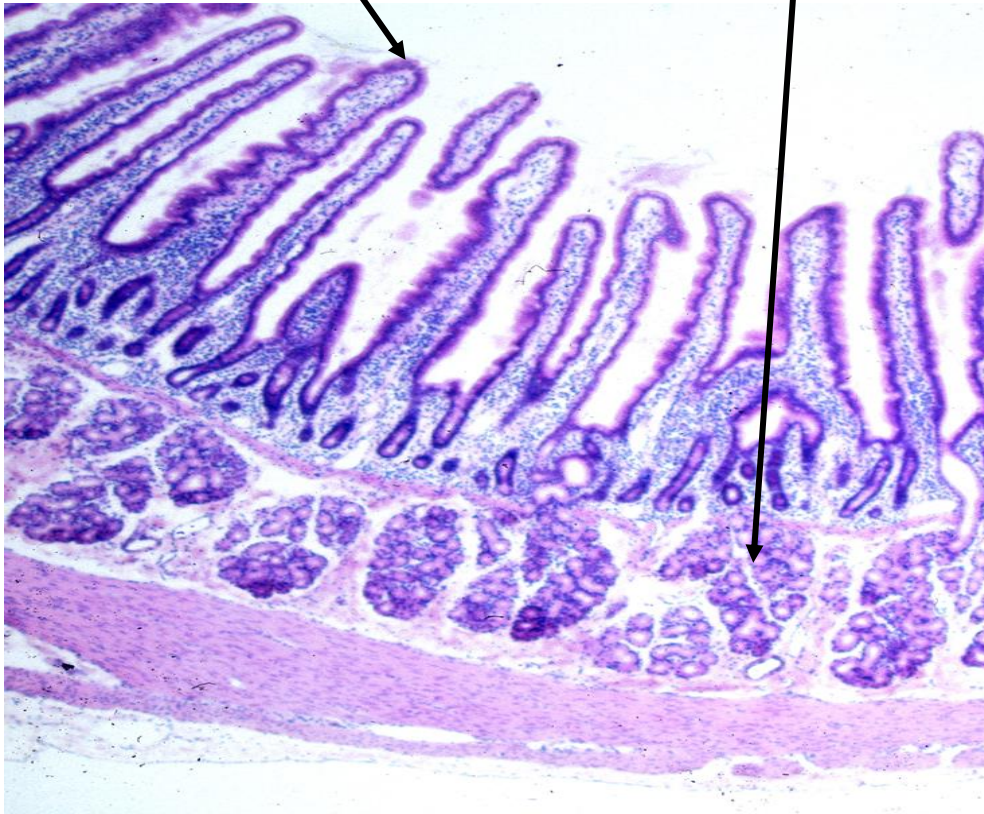
The main functions:

- absorption
- secretion of enzymes

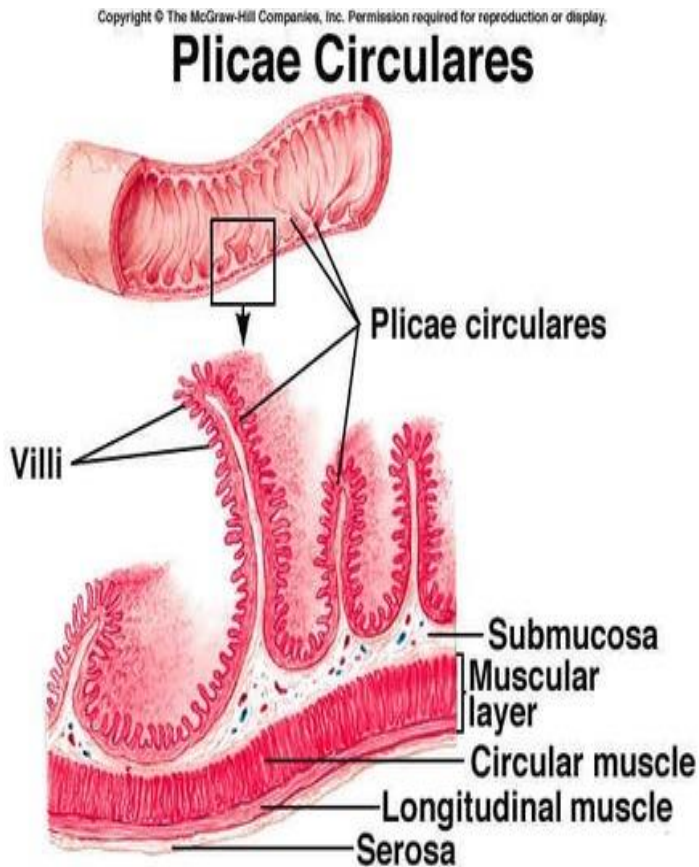


• Duodenum

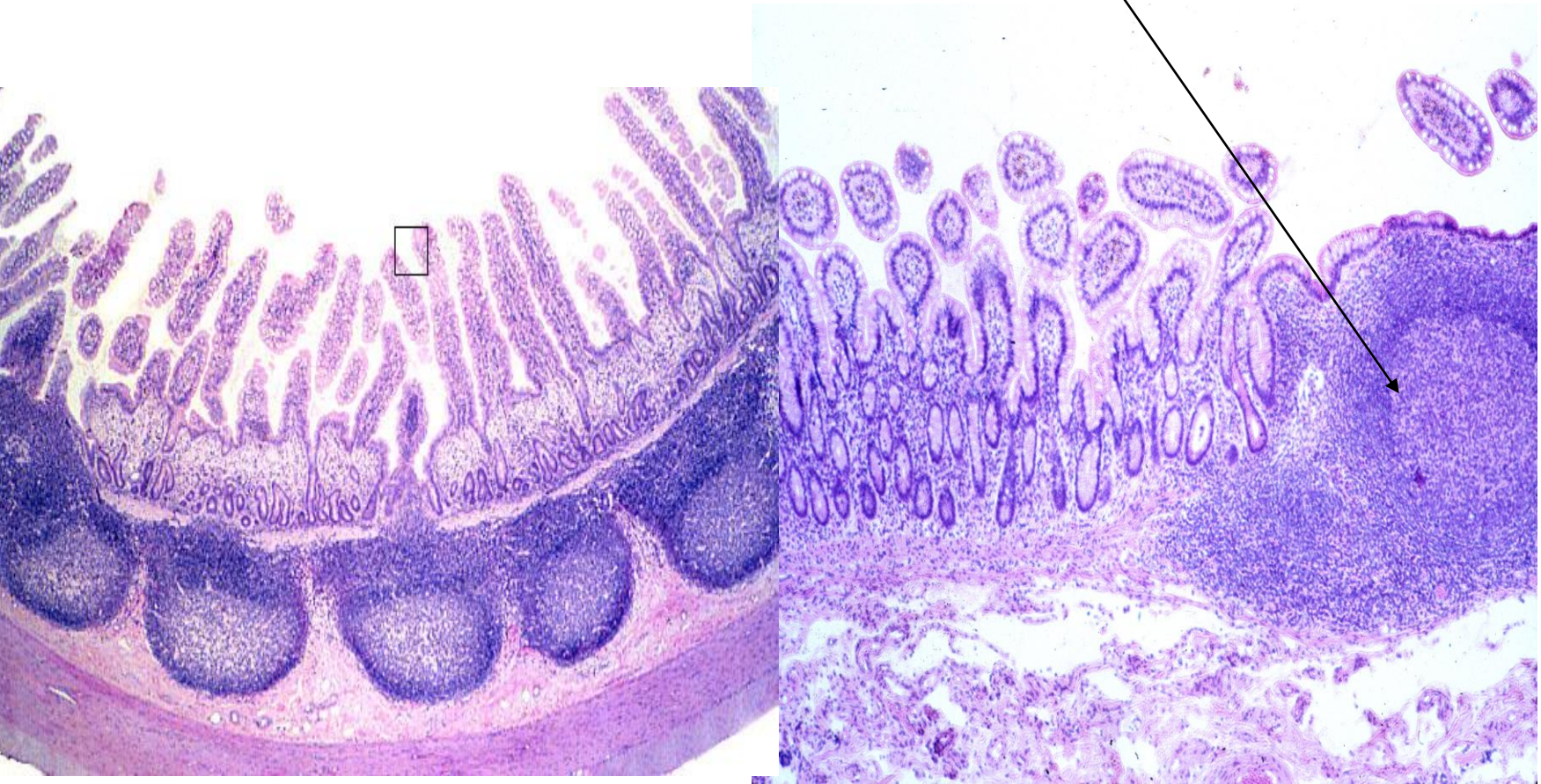
- It has the shape of letter C, bile & pancreatic duct open into it.
- Presence of duodenal(**Brunner's glands**) in the submucosa which secrete alkaline mucus to protect duodenal lining from the acidity of the chyme and bringing the intestinal contents to the optimum pH for pancreatic enzyme action
- has **leaf-shape** villi , few goblet cells , cover by serosa & adventitia



- **Jejunum** : extends from the duodenum to the ileum, has long finger-shaped villi, many plicae circulares , intermediate number of goblet cells .
- Cover by serosa



- **Ileum** : joins the large intestine at the ileocecal valve , fewer club- shaped villi, abundant goblet cells , **Peyer's patches** (lymphoid nodules) are found in the lamina propria of the ileum. Cover by serosa



The wall has the same layers as the rest of tract :

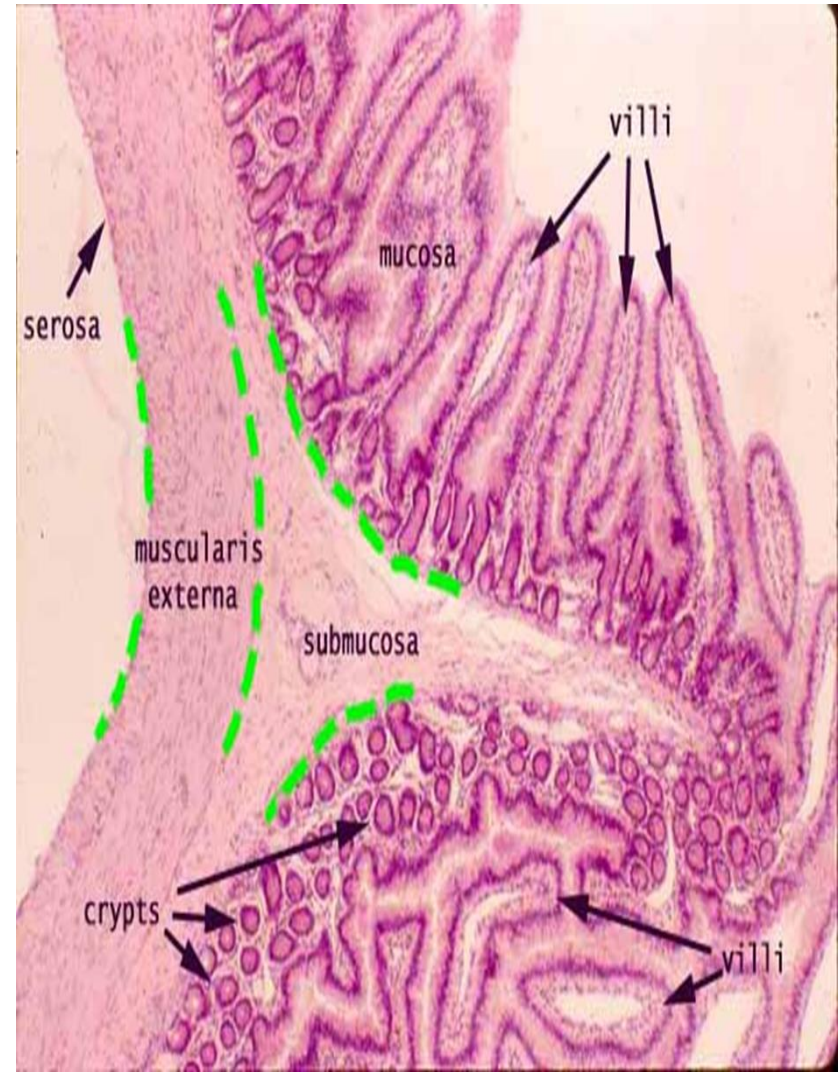
- mucosa
- Submucosa
- Muscularis externa
- Serosa or adventitia

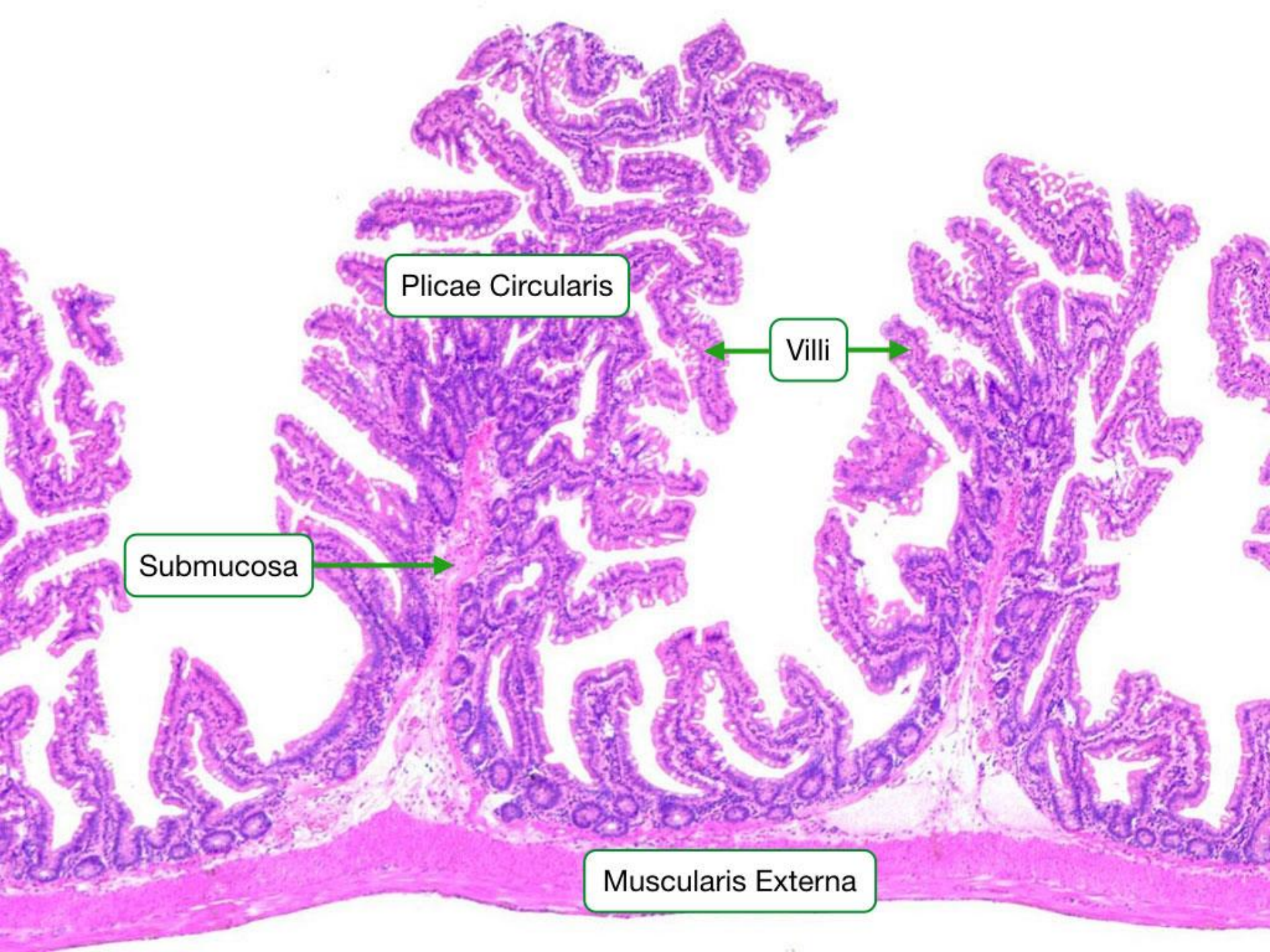
The Mucous membrane

The surface area of the mucous membrane of the small intestine is perform its main function of absorption of food & this is achieved by:

1-The considerable length of the intestine.

- **2-plica circularis:** found in jejunum, consist of mucosa and submucosa. The circular folds tend to slow down the passage of contents through the small intestine, thus facilitating absorption.
- **3-Villi:** Are finger-like processes that project from the surface of the mucosa into the lumen.
- They consist of a core of reticular tissue covered by surface epithelium. The connective tissue core contains numerous blood capillaries and lymphatic vessel
- **4-microvilli:** surface cytoplasm extension





Plicae Circularis

Villi

Submucosa

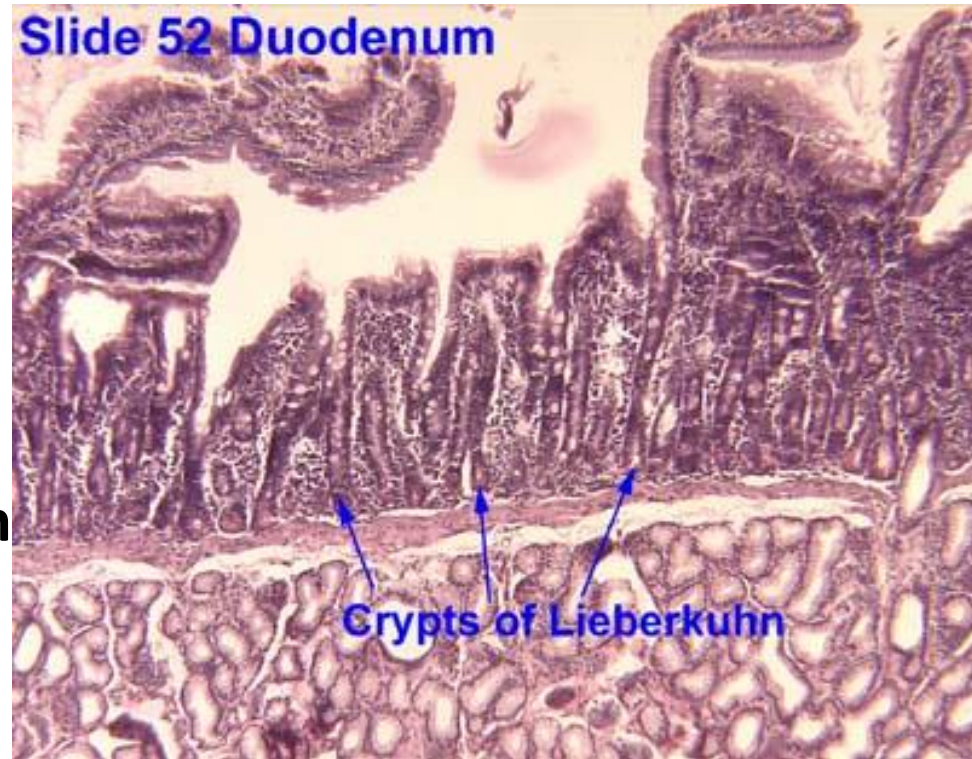
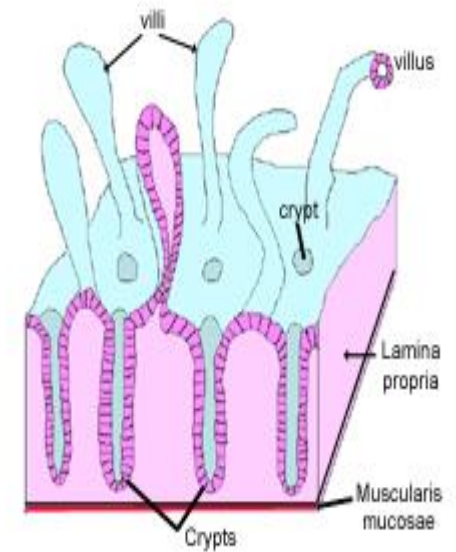
Muscularis Externa

5-Intestinal glands (crypts of lieberkuhn)

Simple tubular glands extend into the lamina propria below the base of villi, they are lined by:

1. -Simple columnar(absorbative cell)
2. -Goblet cell
3. -Paneth cell
4. -Enteroendocrine cell
5. Stem cells
6. M-cells

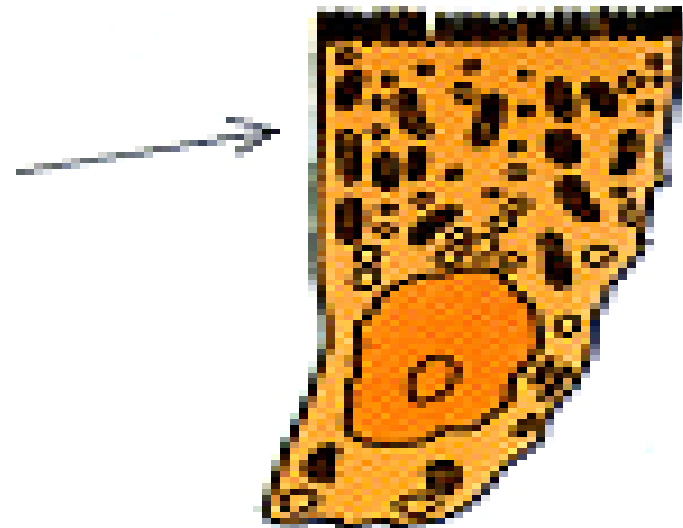
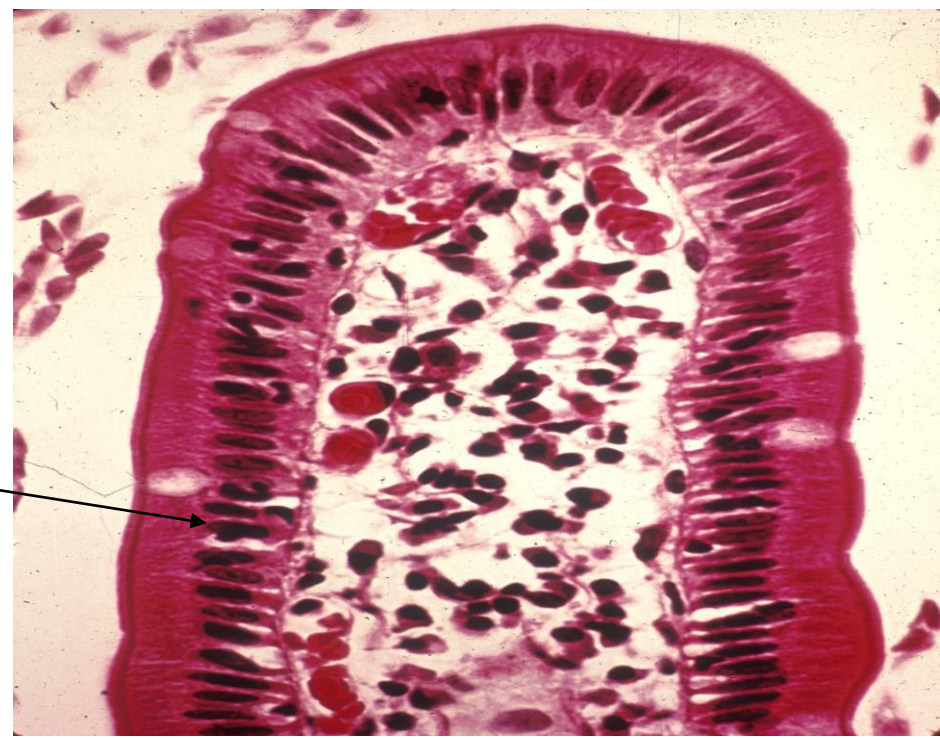
Cells of intestinal crypts secrete intestinal juice, Their secretion enters lumen through small opening between the villi.



Cells cover the surface & gland of the small intestine :

1-absorptive cells

- Are tall columnar
- Oval basal nucleus
- Has brush border are microvilli
- Each cell has ~ 3,000 microvilli for increased surface area

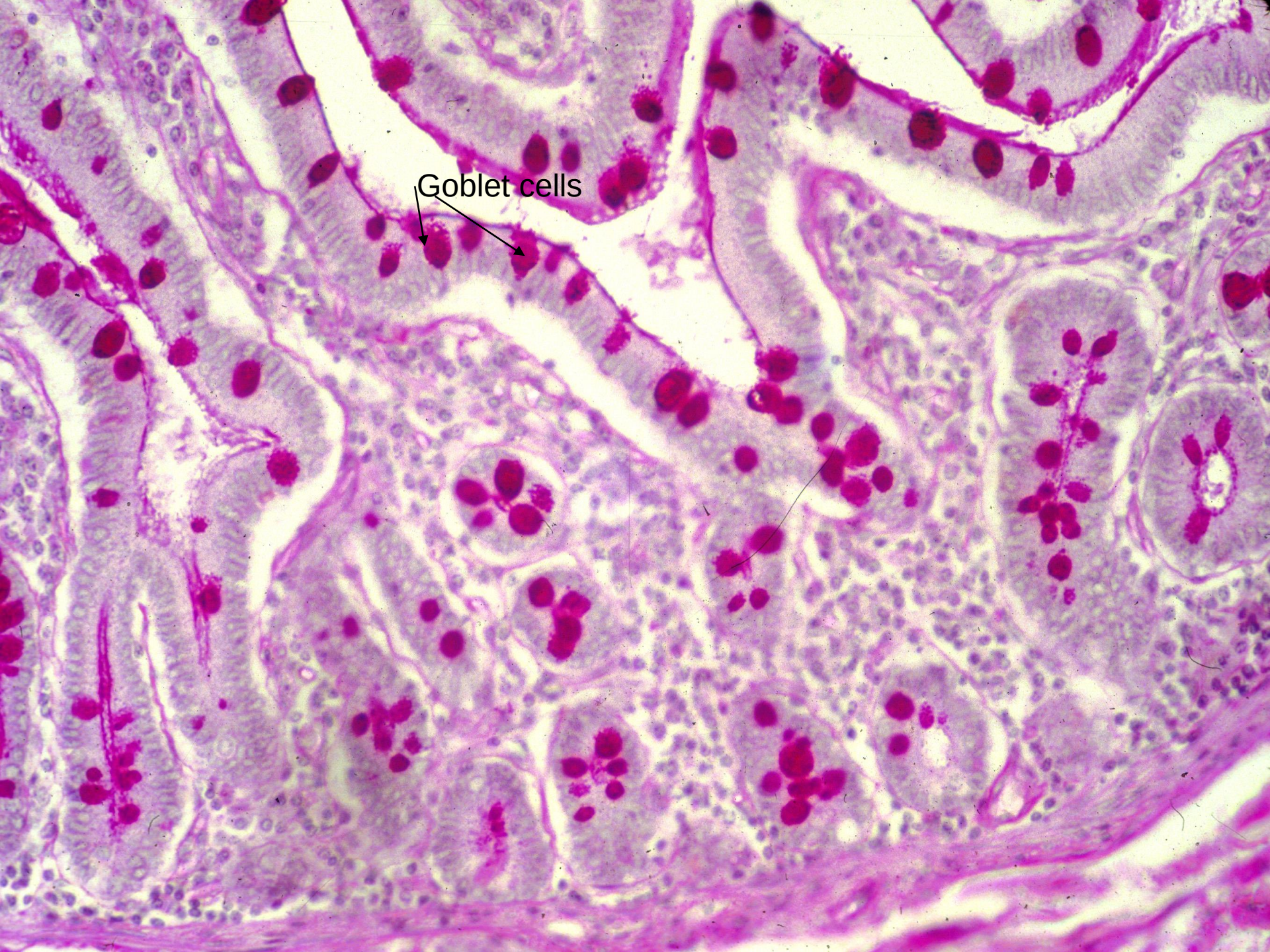


Surface absorptive cell

2-goblet cells: Are interspersed between absorptive cells

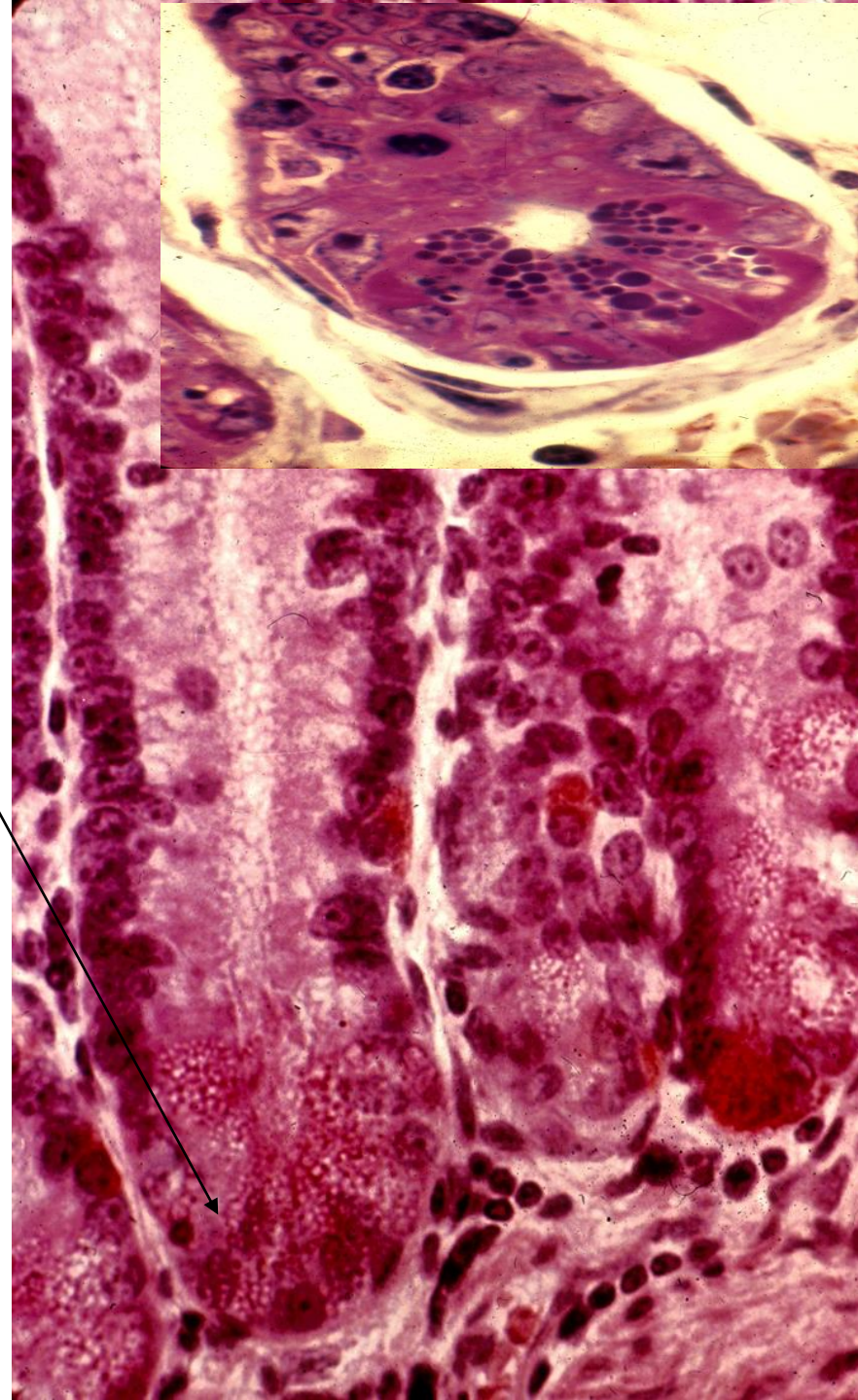
- Increase in the number as they approach the ileum
- Secrete mucus for protection, lubrication





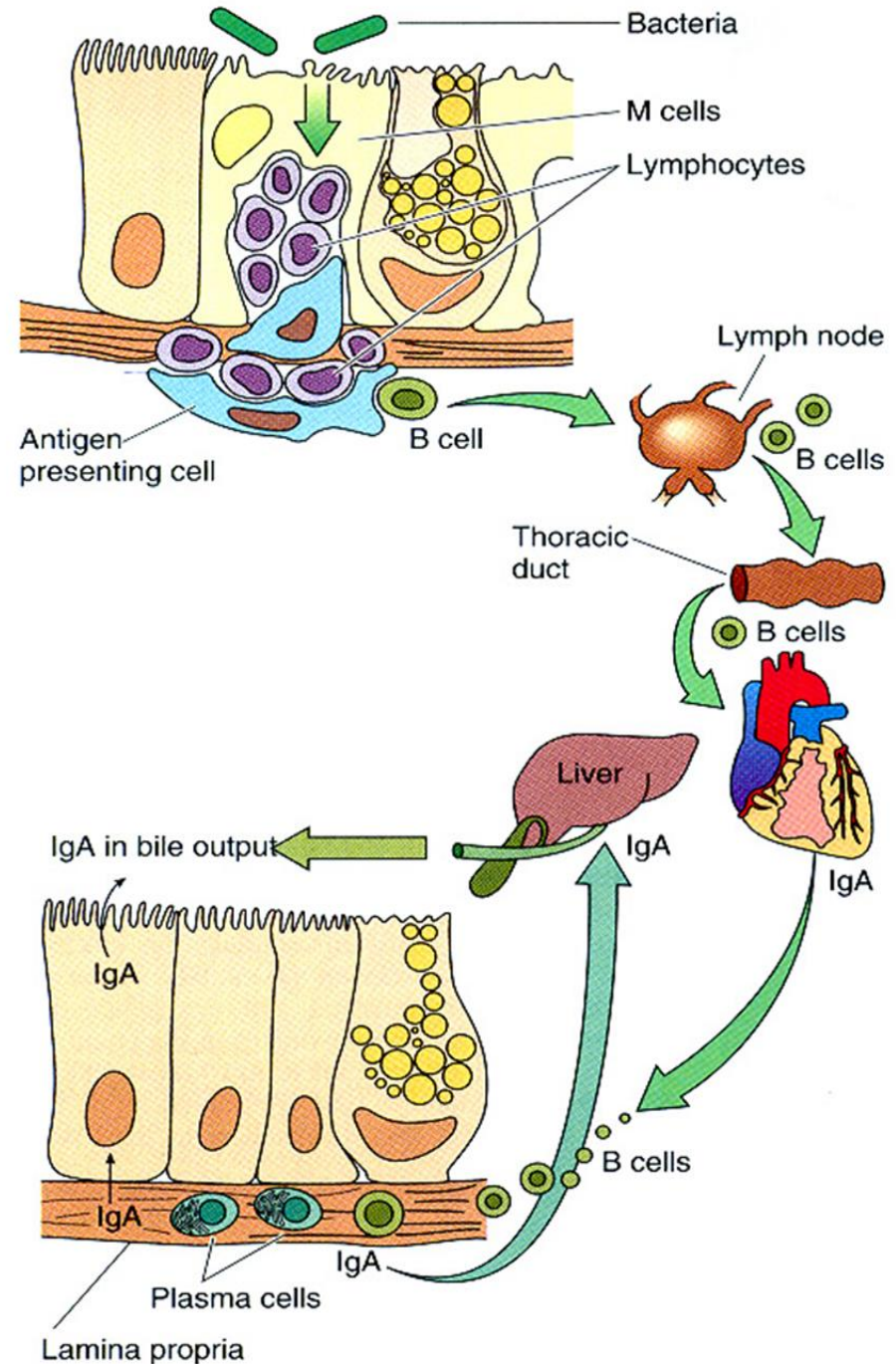
Goblet cells

- **3-paneths cells:**
- Located in the basal portion of intestinal glands
- Are exocrine cells , secretory granules in apical cytoplasm
- Basophilic cytoplasm, eosinophilic granules containing lysozyme
- **Function :** Paneth cell granules release
- lysozyme, phospholipase A2, and hydrophobic peptides called defensins, all of which bind and break down membranes of microorganisms and bacterial cell walls.
- Paneth cells have an important role in regulating the microenvironment of the intestinal crypts.

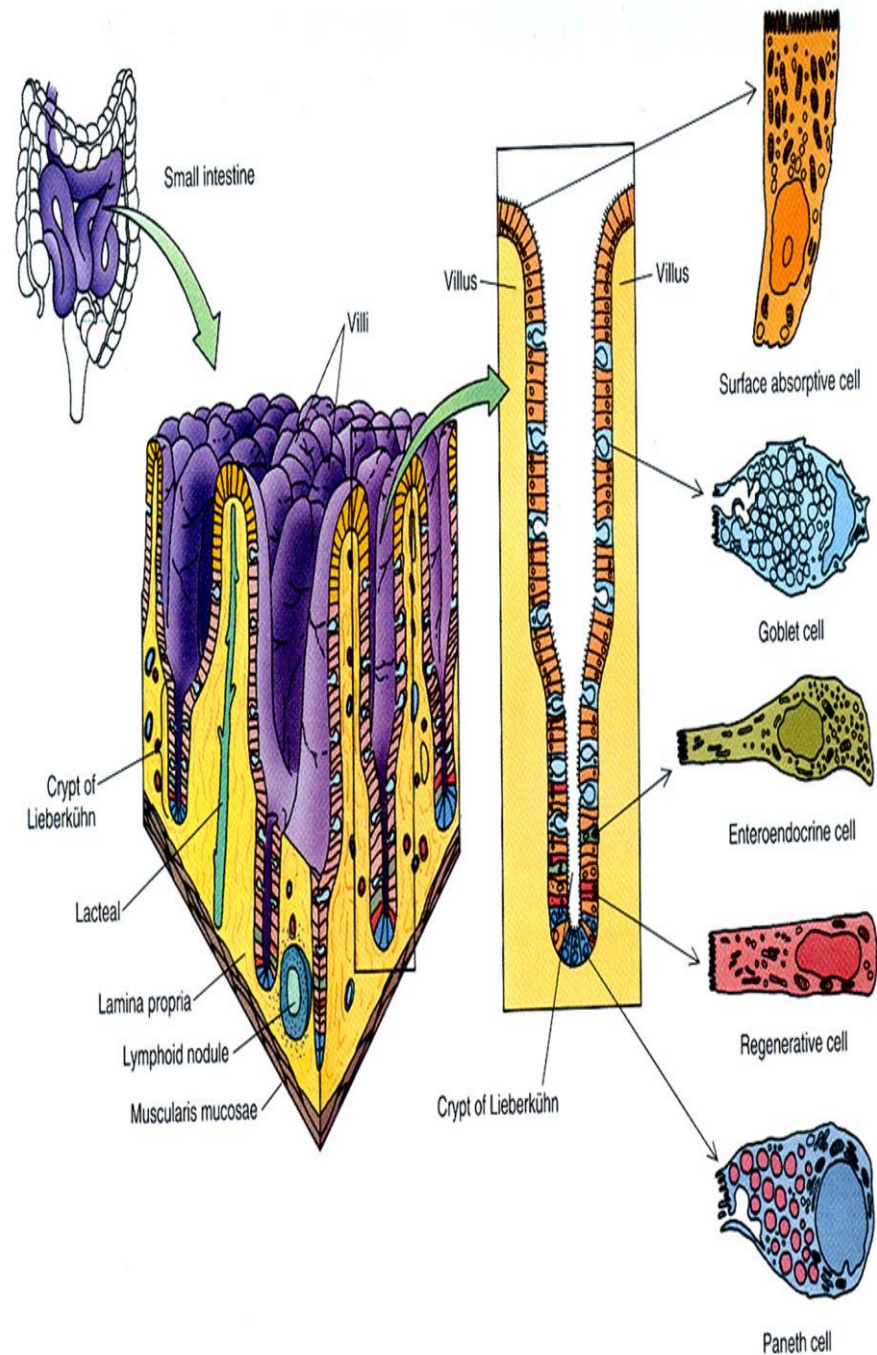


4-M (micro fold) cells:

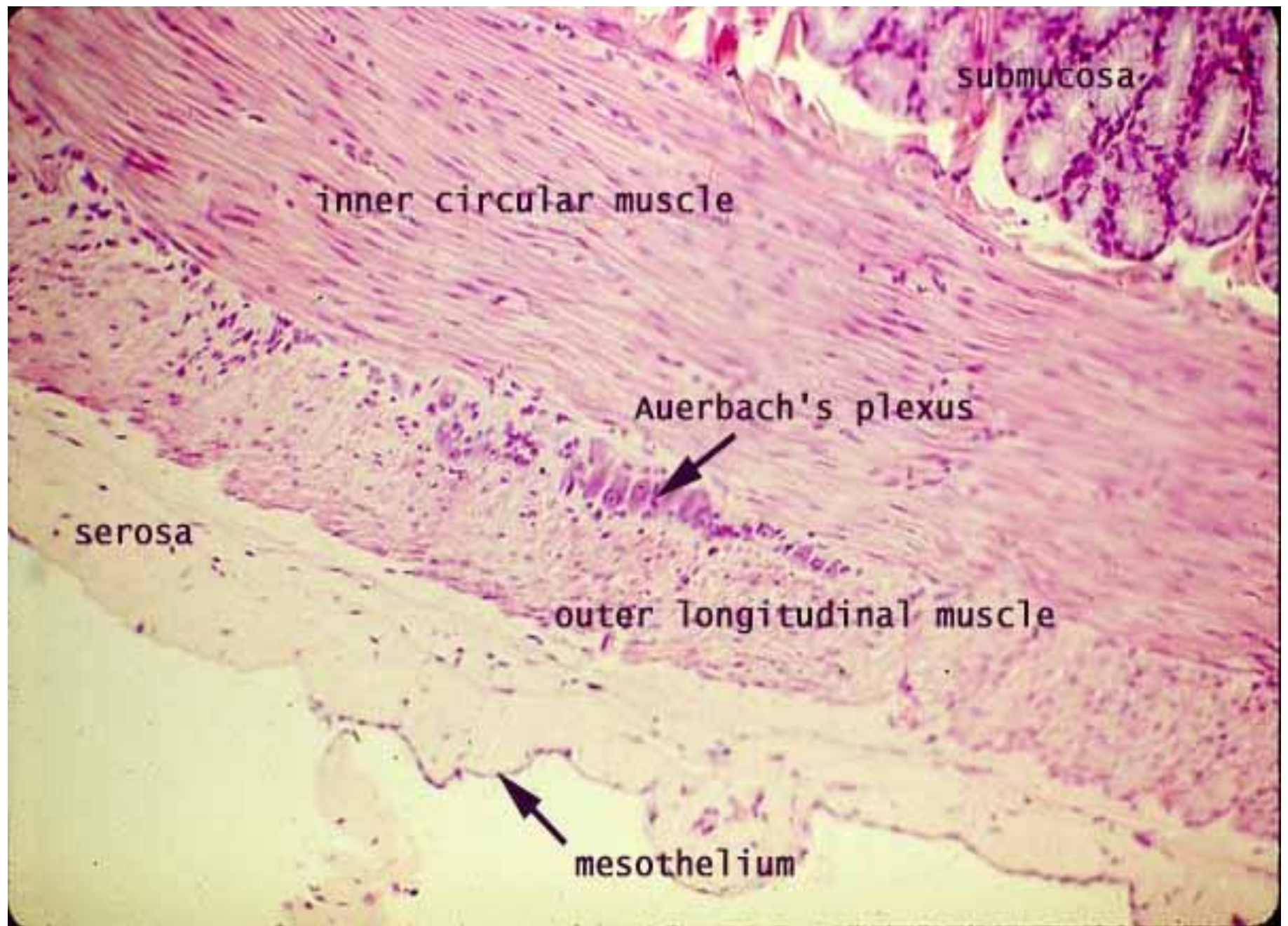
- Are specialized epithelial cells, overlying the lymphoid follicles (peyers patches) in ileum
- Has discontinuous basement membrane
- These cells can endocytose antigens, transport to underlying macrophage, lymphocytes cells
- Representing protective function in intestine



- **5-enteroendocrine cells:**
Located in lower third of crypt resemble those in stomach APUD cells
- Triangular
- Secrete **serotonin, somatostatin, cholecystikinin**
- **peptides, and secretin**
- Regulate intestinal motility and peristalsis, secretions, visceral sensations, and appetite
- **6-stem cells** Located in the base of crypt undergo to mitosis, most GI epithelial cells turn over in 2 - 4 days



- **Lamina propria:** loose c.t. with blood and lymph vessels, nerves, smooth muscle, lymphoid cells and macrophages
- **Muscularis mucosa:** is typical
- **Sub mucosa :** dense c.t. in the duodenum contain Brunner's glands
- **Muscularis externa:** 2 layers inner circular , outer longitudinal
- **Serosa or adventitia**



THANKS